

Hamna Raeel

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Technical Skills

Programming Languages: Python, R, SQL, JavaScript, C++, TypeScript

ML/DL Frameworks: LangChain, LangGraph, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib

Tools & Platforms: HuggingFace, Git, Docker, Kubernetes, CI/CD (Jenkins), Optuna, Weights & Biases

Databases: MySQL, PostgreSQL, Firebase **Cloud:** IBM Cloud, Microsoft Azure, VMware

AI/ML: Model development, EDA, Data preprocessing, Prompt engineering, Model evaluation and deployment, model optimization

Frontend: React, Next.js, Node.js, FastAPI, REST APIs, Tailwind CSS, Redux, Jest, React Native

Work Experience

Artificial Intelligence Diagnostic Lab, FAST-NUCES

Islamabad

Research Assistant

Oct 2025 – Jan 2026

- Developed and trained deep learning segmentation models using Mask R-CNN, Fast R-CNN, SegFormer, and DETR.
- Optimized preprocessing, augmentation, and training pipelines to improve histopathology image segmentation performance.
- Built and evaluated deep learning classification models using ResNet, DenseNet, EfficientNet, Vision and Swin Transformer.
- Improved histopathology disease classification through feature extraction, model optimization, and performance evaluation.

Software Productivity Strategists, Inc. (SPS)

Islamabad

DevOps and Cloud Infrastructure Intern

Jul 2025 – Sep 2025

- Developed ELT data pipelines and automated CI/CD workflows using Jenkins for reliable deployments.
- Containerized applications with Docker and orchestrated deployments using Kubernetes on IBM Cloud.
- Implemented monitoring and logging using Prometheus and Grafana for cloud-native applications.
- Managed deployments and infrastructure across multi-cloud environments, including Azure and IBM Cloud.

Artificial Intelligence Diagnostic Lab, FAST-NUCES

Islamabad

Data Science Intern

Mar 2025 – May 2025

- Developed and evaluated ML/DL models using CNN, ResNet, UNet, SVM, and Decision Trees across benchmark datasets.
- Applied ML to medical imaging and retail datasets, including CBIS-DDSM, Chest X-Ray, Retinal Vessel, and CIFAR-10/100.
- Implemented RAG using vector databases, embedding models, and LLMs for intelligent document search and question answering.

Optimus Bee

Lahore

Software Engineer

Nov 2022 – Dec 2024

- Built an e-commerce platform with integrated Stripe payment processing, secure authentication, and order management.
- Developed a Mental Health Wellness application with an AI-powered chatbot, mood tracking, and recommendations.
- Created a Financial Budget Management Tool for expense tracking, categorization, and analytics dashboards.
- Built an admin dashboard with dynamic UI, REST APIs, user management, and reporting for an Australian travel agency.

Cyber Legends

Lahore

Front End Developer

Jun 2021 – Oct 2022

- Developed a university portal website for NASTP-IIT using WordPress and custom themes.
- Built "Infinite Writing," a freelance services platform with dual dashboards using HTML, Tailwind CSS, and JavaScript.

Research Projects

Segmentation-Guided Histopathology Analysis for Hirschsprung Disease Detection

- Developed a Mask2Former-based segmentation model with ResNet–Swin Transformer backbone, achieving 92% Dice Score, followed by a classification framework that achieved 94% accuracy and 95.3% AUC-ROC on histopathology images.

Meta-Learning for Zero-Shot Cross-Lingual Transfer (CML-mDeBERTa)

- Implemented a Curriculum Meta-Learning framework with first-order MAML on the XNLI dataset across 15 languages, achieving 80.34% mean accuracy and an 11% improvement over baseline for zero-shot cross-lingual transfer.

Sentence Classification using Convolutional Neural Network

- Designed and implemented a TextCNN model for sentence classification on the AGNews dataset, achieving high performance accuracy using random, static, non-static and multichannel word embeddings.

Education

FAST, National University of Computer and Emerging Sciences (NUCES)

Islamabad

Master of Science in Data Science

- Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI

Certifications & Trainings

Google Advanced Data Analytics (Google); Databases and SQL for Data Science with Python (IBM); ChatGPT Prompt-Engineering for Developers (DeepLearning.AI); AI concepts for developers and technology professionals (Microsoft); Get started with AI applications and agents on Azure (Microsoft)